Kirtan Patel

Chicago, IL | +1 (773) 701-0710 | kpate452@uic.edu | linkedin.com/in/patelkirtan13 | github.com/kirtanpatel2003

Summary

Graduate Computer Science student with hands-on experience in full-stack development, IT operations, and technical support. Skilled in JavaScript, ReactJS, Node.js, Python, and RESTful services, with a background spanning software engineering, helpdesk and tier-2 support, network setup, hardware repair, and team leadership. Experienced in managing IT infrastructure, building internal tooling, and training technical staff in university environments.

Education

University of Illinois Chicago (UIC) — Chicago, IL
Master of Science in Computer Science — *May 2027 (Expected)*

- Relevant Coursework: Network Security, Cryptography, Computer System Security, Wearable Technology, Data Visualization & Analytics

University of Illinois Chicago (UIC) — Chicago, IL
Bachelor of Science in Computer Science – Software Engineering Concentration; Minor in Mathematics — *May 2025*

- Dean's List: College of Engineering – Fall 2023 & Fall 2024
- Relevant Coursework: Building Secure Computer Systems, Artificial Intelligence, Database Systems, Software Engineering 1 & 2, UI/UX, Software Design, Statistical Methods, Data Structures

Skills

Languages: Python, JavaScript, HTML5, CSS3
Frameworks & Libraries: ReactJS, Node.js, Express.js, Flask, Next.js
Databases: MongoDB, MySQL, PostgreSQL
Tools & Platforms: Git/GitHub, Docker, Postman, Vercel, Linux
Hardware & Fabrication: Arduino, Processing, 3D Printing

Work Experience

Technology Solutions, Office of the Vice Chancellor for Innovation – UIC — Chicago, IL
IT Helpdesk Support Manager — *Dec 2024 – Present*

- Serve as tier-2 helpdesk support, handling escalated tickets in TeamDynamix (TDX) covering classroom network setup, AV systems, and campus-wide technology issues.
- Diagnose and repair laptops and client devices; provide direct support to faculty, staff, and students across campus.
- Interviewed 8 of 21 candidates and authored troubleshooting documentation adopted across the support team.
- Manage 30 printers across East Campus; streamlined operations to 24 active units, reducing average downtime from 6 units to 1 and saving thousands annually.

Technology Solutions, Office of the Vice Chancellor for Innovation – UIC — Chicago, IL
Classroom Technology Support Specialist — *Aug 2022 – Dec 2024*

- Assisted professors with classroom equipment issues including projectors, audio systems, and lecture capture setups to ensure uninterrupted instruction.
- Set up and configured computer labs for courses and campus events.
- Trained and mentored incoming support specialists, building their ability to independently resolve classroom technology issues.
- Provided printer support and maintenance across assigned campus locations.

Projects

TrailBack | *Arduino, ReactJS, BoxMap API, BMP280, Magnetometer, MPU6050, MQ135*

- Building a GPS-free navigation device for trekkers using sensor fusion across a pressure sensor, magnetometer, and accelerometer-gyroscope to calculate heading, altitude, and displacement. Online mode takes a starting coordinate and uses BoxMap API to anchor the start point on a map; offline mode runs with no coordinate or internet at all.
- All path tracking in both modes is driven purely by sensor data. Built a ReactJS web interface with a live Arduino data feed to display real-time navigation and return-path guidance.

Mini Tor Anonymity Network | *Python, TCP Sockets, Curve25519, AES-256-GCM, HKDF, SHA-256*

- Implemented Tor's three-hop onion routing protocol from scratch as six independent processes (directory, guard/middle/exit relays, client, destination) communicating over real TCP, with telescoping EXTEND cells so the client only ever connects to the guard while middle and exit are reached through the growing circuit.
- Built the entire cryptographic stack in pure Python with no third-party crypto libraries: SHA-256 (FIPS 180-4), HMAC, HKDF (RFC 5869), X25519 Diffie-Hellman via Montgomery ladder (RFC 7748), AES-256 (FIPS 197), and AES-256-GCM (NIST SP 800-38D), each verified against published NIST and IETF test vectors.

WeatherGenie | *ReactJs, OpenWeather API, Arduino*

- Engineered an interactive weather forecasting platform integrating the OpenWeather API to deliver real-time, location-aware weather updates via a responsive landing page.
- Combined Arduino-powered local sensor data with API-based global forecasts to deliver room-level precision for temperature, humidity, and environmental analytics with interactive visualizations.

FaceWay | *ReactJs, Express.js, face-api.js, TensorFlow.js*

- Built a kiosk application with face recognition login using face-api.js and TensorFlow.js, eliminating manual input by automatically identifying users and retrieving their profiles on approach.
- Integrated a personalized interface that dynamically adjusts based on user history and displays account-linked points and promotional offers in real time.

KeyTracker | *Next.js, MongoDB, Vercel*

- Built and deployed a barcode-based key checkout system for UIC Technology Solutions, enabling staff to check keys in and out with real-time availability tracking.
- Implemented audit logging, conflict detection (409 responses for double-checkouts), and mismatch flagging to enforce access control integrity and maintain a reliable activity trail.

DiffFence | *Python, OSV.dev, Ollama*

- Built a local-first Python CLI that acts as a diff-aware security gate, scanning code changes for security vulnerabilities and production risks before a push or merge.
- Implemented baseline-on-first-run logic so subsequent scans surface only new findings, eliminating repeated noise and keeping developer focus on incremental risk introduced by each change.

Email Summarizer | *Python, Ollama, Telegram API*

- Built a Python tool that fetches unread emails, summarizes content using a locally-run Mistral model via Ollama, and assigns priority levels to each message.
- Delivers prioritized summaries directly to the user through a Telegram bot, reducing inbox overhead without relying on cloud-based LLM services.

Certifications

ChatGPT Prompt Engineering for Developers — May 2024
OpenAI – DeepLearning.AI

Human Research – Social & Behavioral Research — Sep 2024
CITI Program – University of Illinois Chicago

Foundations of Cybersecurity — Oct 2024
Google – Coursera

Introduction to Networking — Oct 2024
NVIDIA – Coursera